MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) FIRST SEMESTER EXAMINATION (2024 ADMISION ONWARDS)

Semester- 3

MG3DSEPSY200- BIOPSYCHOLOGY

Time: 2 hrs **Marks: 70**

Part A: Answer all questions. Each carry 1 mark.

1. Which of the following methods involves destroying brain tissue to study behavior?

A. EEG B. Lesioning C. MRI D. EMG

(CO1, U)

2. The main function of the sympathetic nervous system is to:

A. Slow down heart rate
B. Stimulate digestion
C. Prepare the body for 'fight or flight'
D. Regulate body temperature

(CO3, An)

3. Which neurotransmitter is primarily involved in muscle movement?

A. Dopamine B. Serotonin C. Acetylcholine D. GABA

(CO2, An)

4. The left hemisphere of the brain is generally associated with:

A. Spatial ability
B. Musical ability
C. Language and logic
D. Emotional processing **(CO2, An)**

5. Which endocrine gland regulates the body's metabolism?

A. Pituitary gland B. Thyroid gland C. Adrenal gland D. Pineal gland

(CO3, U)

6. The Hypothalamic-Pituitary-Adrenal (HPA) axis is primarily involved in:

A. Sexual behavior
B. Visual processing
C. Stress regulation
D. Hunger management **(CO3, An)**

7. The 'reward system' in the brain is majorly linked to which structure?

A. Medulla
B. Amygdala
C. Hippocampus
D. Nucleus Accumbens **(CO2, An)**

8. The action potential in a neuron is initiated when:

A. Synaptic vesicles are released B. Sodium ions rush inside the neuron
C. Potassium ions move inward D. Neurotransmitters are absorbed

(CO2, U)

9. Which scanning technique uses magnetic fields and radio waves?

A. PET B. MRI C. CT D. EEG

(CO1, U)

10. Which hormone is released during stress and helps maintain energy supply?

A. Oxytocin B. Cortisol C. Insulin D. Prolactin

(CO3, An)

11. Functional asymmetry in the brain refers to:

A. Equal processing on both sides
B. Dominance of one hemisphere for specific tasks
C. Brain degeneration
D. Involuntary muscle movements

(CO2, An)

12. Which of the following neurotransmitters is classified as inhibitory?

A. Acetylcholine B. Glutamate C. GABA D. Serotonin

(CO2, U)

13. Which brain region plays a central role in regulating hunger and thirst?

A. Hippocampus B. Hypothalamus C. Cerebellum D. Prefrontal cortex

(CO5, An)

14. The main function of the limbic system is:

A. Coordinating voluntary movement
B. Processing emotions and memory
C. Controlling hormone secretion
D. Maintaining balance

(CO4, An)

15. The neural mechanism of sexual behavior is primarily associated with which part of the brain?

A. Brainstem B. Hypothalamus C. Cerebellum D. Corpus callosum

(CO4, An)

16. Which feedback mechanism reduces the output or activity of any organ or system back to its normal range?

A. Positive feedback B. Circular feedback C. Negative feedback D. Reflex arc

(CO3, U)

17. The ethical principle in biopsychological research requires:

A. Avoiding animal models B. Deceiving participants
C. Protecting participants from harm D. Disclosing hypotheses to subjects

(CO1, U)

18. Which hormone plays a role in bonding and attachment behavior?

A. Testosterone B. Oxytocin C. Adrenaline D. Cortisol

(CO3, An)

19. A disorder of excessive eating is best categorized under:

A. Sleep disorders B. Learning disorders C. Eating disorders D. Hormonal disorders

(CO5, An)

20. Which technique is most useful for recording brain's electrical activity?

A. CT B. MRI C. EEG D. PET

(CO1, U)

(20*1= 20 Marks)

Part B: Answer any 10 questions. Each question carries 3 marks

21. Define biopsychology and explain its relationship with neuroscience. **(CO1, U)**

22. List any two ethical principles to be followed in biopsychological research. **(CO1, U)**

23. Describe the process of synaptic transmission in simple terms. **(CO2, U)**

24. Identify any two major regions of the brain and state their general functions. **(CO2, U)**

25. Differentiate between the sympathetic and parasympathetic nervous systems. **(CO2, An)**

26. What is brain lateralization? Give one example. **(CO2, U)**

27. Name any two major endocrine glands and mention one hormone each produces. **(CO3, U)**

28. Explain the role of the HPA axis in stress regulation. **(CO3, An)**

29. How do hormones influence learning and memory? **(CO3, An)**

30. Mention any two neural mechanisms involved in sexual behavior. **(CO4, An)**

31. What is the role of the hypothalamus in emotional behavior? **(CO4, An)**

32. List any two physiological factors that influence eating behavior. **(CO5, An)**

33. Identify the brain mechanism involved in thirst regulation. **(CO5, An)**

(10*3= 30 Marks)

Answer any 2 questions. Each carry 10 marks

34.Describe the scope and methods of study in biopsychology.Explain the ethical considerations in biopsychological research. **(CO1, U)**

35. Explain the process of nerve impulse transmission and the role of neurotransmitters. **(CO2, An)**

36. Discuss the role of hormones in stress and emotional regulation.Explain the mechanisms of the HPA and HPT axes in hormonal feedback. **(CO3, An)**

37. Examine the physiological basis of sexual behavior and emotionality. **(CO4 & CO5, An)**

(10*2= 20 Marks)